Experiment 20
Ohm's Law

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Abstract

This experiment investigates the characteristics of resistors arranged in series, parallel, and mixed circuits. The main objective is to demonstrate how current, voltage, and resistance behave differently in each configuration, but always verifying Ohm's Law. To do so, different circuits are constructed, measuring its characteristics and testing if they are close to the theoretical predictions. In addition, the linear relationship between voltage and current for a constant resistance, studied in the final part of the experience, confirms Ohm's Law. The results show a strong agreement between measured and expected values.

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1 Introduction

1.1 Contextualization

In the XIX century the study of the electric phenomena reached its peak, attracting the attention of important scientists such as André-Marie Ampère, Michael Faraday, James Clerk Maxwell and Georg Simon Ohm. Their investigations contributed to the development of more complex technologies, improving our quality of life and, therefore, supporting the prosperity of humanity.

This experiment focuses on Ohm's Law, which states that the voltage V across a conductor is directly proportional to the current I flowing through it, with the resistance R as the proportionality constant. Mathematically, this law is expressed as $V = IR$. This simple but powerful equation laid the foundation for other electrical theories.

This experience aims to verify Ohm's Law and explore the basic rules that govern the combination of resistors in series and in parallel.

1.2 Theorical principles

In the context of an electric circuit, many physical magnitudes can be defined, for instance:

- the voltage V, measured in volts, quantifies the difference of the electric potential between two points of the circuit
- the intensity I refers to the charge that flows through per time, i.e., $I = \frac{dQ}{dt}(A)$
- the resistance R, which can be described as the opposition to the flow of electrons through the circuit. Its unit is the ohm, Ω.

These parameters are related by Ohm's Law, which is expressed as follows.

$$V = R \cdot I \tag{1}$$

As the main objective of this experimental practice is to verify the validity of equation 1, we will design and assemble a variety of electric circuits that include resistors with known values. To achieve this, we will utilize two fundamental methods of resistor configuration: series and parallel associations. These types of associations are illustrated in Figures 1 and 2, respectively.

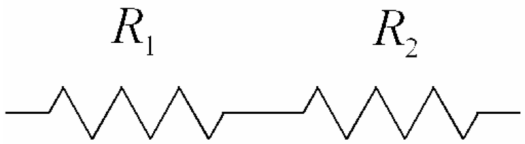


Figure 1: Electric circuit with two resistors associated in series.

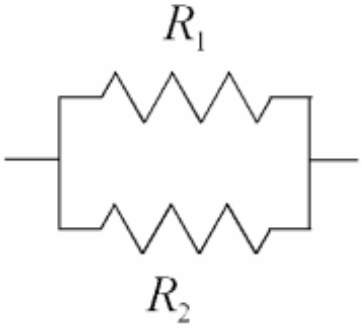


Figure 2: Electric circuit with two resistors associated in parallel.

It is important to understand how the total resistance behaves under different configurations. In the case of series connection the current flows through each resistor sequentially, resulting in the total resistance being equal to the sum of the individual resistances, as described by the equation:

$$R_T = R_1 + R_2 + ... + R_N \ (\Omega). \tag{2}$$

$$V_T = V_1 + V_2 + ... + V_N \quad and \tag{3}$$

$$I_T = I_1 = I_2 = ... = I_N. \tag{4}$$

On the other hand, in a parallel configuration, the resistors are connected across the same two points, meaning that the voltage across each resistor is the same (6). However, the total current flowing through the circuit is divided among the parallel branches (7). The effective total resistance in this type of configuration is determined by the formula:

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + ... + \frac{1}{R_N} \ (\Omega). \tag{5}$$

$$V_T = V_1 = V_2 = ... = V_N, \tag{6}$$

$$I_T = I_1 + I_2 + ... + I_N. \tag{7}$$

These formulas are quite helpful when analyzing more complicated circuits, as they allow us to simplify sections of a circuit by reducing a group of resistors, either in series or in parallel,to an equivalent resistance R_T. This process is schematized in the following image.

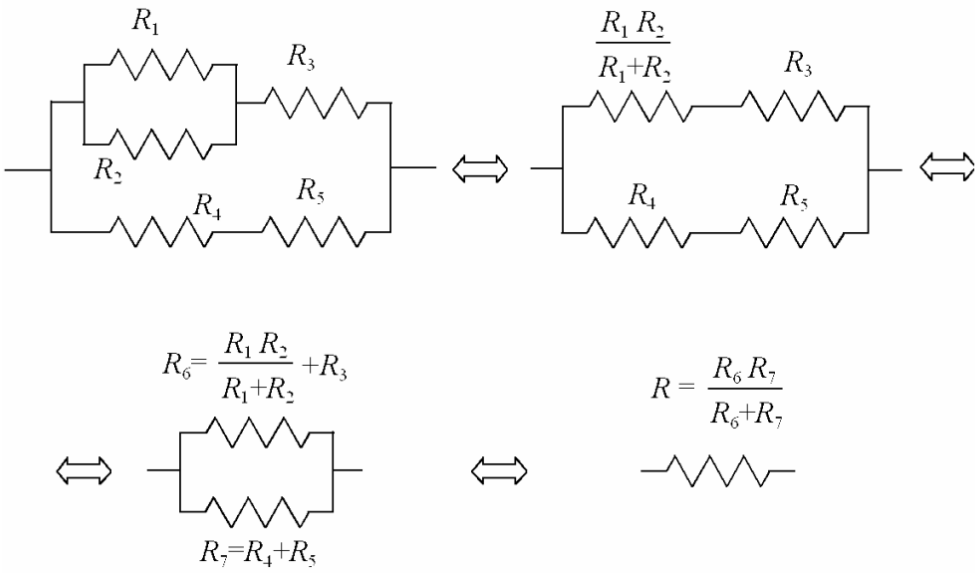


Figure 3: Circuit with multiple resistors simplified into one with only one resistor.

It is relevant to note that Figure 1, 2 and 3 were extracted from the Laboratory Manual [1].

2 Materials and Methods

For this practice, the following materials were used [1].

2.1 Materials

- **Multimeter**. It is an electronic device used to measure parameters in electrical circuits -for example: voltage, current, resistance, frequency and capacitor capacitance-, both for DC and AC.
- **Mounting plate**. A mounting plate is a flat and sturdy surface used to construct electrical circuits in a secure manner.
- **Wires**. They are used to connect different components set up on the mounting plate.
- **DC source.** It is a device that provides a direct current (DC) electrical output, which means that the current flows in one direction only.
- **Resistors.** They are commonly used to control the amount of current that flows through a circuit. In this practice we used five resistors of 470, 1000, 2200, 3300 and 4700 Ω.
- **Bridge connector**. It is a component used to link two points in a circuit without modifying the current I, which means it nearly has resistance.

2.2 Methods

2.2.1 Resistance measurements

First of all, we need to choose five resistors, preferably with a value between 500 and 5000 Ω and enumerate them from one to five, which we did in increasing order.

Once we have chosen the resistors and compared the manufacturer value with the one we obtained directly measuring each resistor with the multimeter, we have to set up the following circuits.

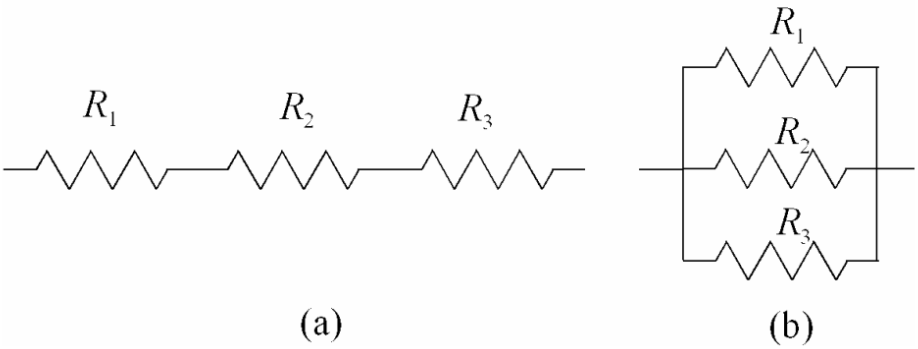


Figure 4: Simple circuits with resistors associated in series (a) and in parallel (b).

We will measure the equivalent resistance of circuit (a) and (b) and compare it with its theoretical value, which can obtained from equations 2 and 3 respectively. In addition, we are going to carry out the circuit described in Figure 3 and, again, compare its theoretical value with the one measured with the multimeter.

2.2.2 Voltage measurement

Then, using circuits (a) and (b) from Figure 4 and a DC source, we will set up the following circuits.

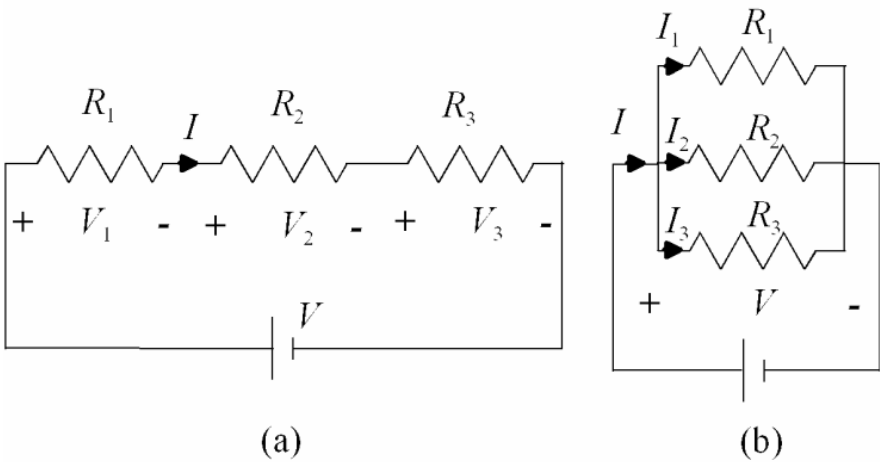


Figure 5: Simple circuits with resistors associated in series (a) and in parallel (b), connected to a DC source.

For the first circuit, we need to measure V_1, V_2, V_3 and V_T, proving that the relation 3 is fulfilled; as well as testing that the current is the same in every part of the circuit (4).

Regarding the second circuit, the objective is to check the veracity of the formula 7, measuring I_1, I_2, I_3 and I_T, and comparing these values with its theoretical numbers, which can be calculated with Ohm's Law. In addition, it is relevant to verify relation 6.

2.2.3 Ohm's Law testing

In order to prove Ohm's Law (1), we are going to set up the following circuit.

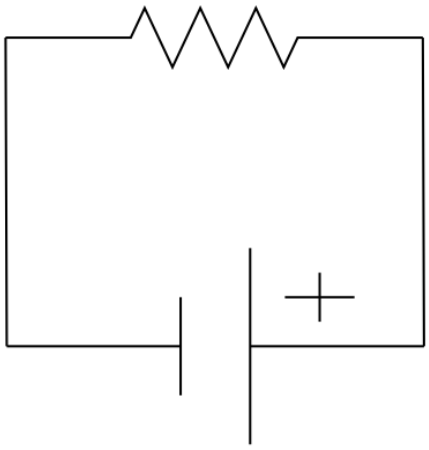


Figure 6: Simple circuit with a resistor R, chosen from the five we have, and connected to a DC source.

Once constructed, we will change the voltage supplied from 1 to 10 volts, increasing one volt each time. For each voltage, we need to measure with the multimeter the actual voltage V across the resistor and the current I flowing through it. We will take note of this measurements, as well as the range of the multimeter used, and organize the data in a table.

Then, we are going to plot a graph using the least squared method, with the current I on the horizontal axis and voltage V on the vertical axis. This graph should be a straight line, as stated in equation 1, whose slope corresponds to the resistance R.

Finally, we will compare the data obtained from the graphic with the manufacturer value for R and its uncertainty.

As well as in the theorical principles' section, images 4, 5 and 6 were extracted from the Laboratory Manual [1].

3 Results and discussion

The results obtained in this experiment can be divided in the next sections.

3.1 Resistor measurements

3.1.1 Individuals

Firstly, we have examined the resistance values measured using our precision multimeter, including their associated uncertainties, and compared them with the manufacturer's specifications. The experimental data and comparative analysis are presented in the next table.

Resistors	Expected R ($K\Omega$)	Tolerance 5% ($K\Omega$)	Experimental R ($K\Omega$)	$u_C(R)$ ($K\Omega$)
R_1	0,470	0,023	0, 4700	0,0068
R_2	1,000	0,050	1,002	0,011
R_3	2,20	0,11	2,170	0,047
R_4	3,30	0,17	3,340	0,057
R_5	4,70	0,24	4,730	0,068

Table 1: Measured values, nominal values, percentage deviations, and measurement uncertainties for each resistor

As the table shows, the measured values of each resistor are very close to those provided by the manufacturer.

3.1.2 Series

If we distribute the resistors so that they are in series, the expected total resistance will be $R_T = 3,642 \pm 0,049$ ($K\Omega$) which, compared to the experimental measurement, $R_T = 3,64 \pm 0,059$ ($K\Omega$), confirms that the equation describing this total resistance (equation 2) is accurate.

On the other hand, using the DC source, we have taken the voltage corresponding to each resistor, as well as the voltage that circulated through the complete circuit. Similarly, the current flowing through each resistor and the total current of the circuit were measured.

Voltage	Experimental V (V)	$u_C(V)$ (V)
V_1	1,0450	0,0082
V_2	2,220	0,041
V_3	4,820	0,054
V_T	8,110	0,071

Table 2: Individual voltages where V_1, V_2 and V_3 correspond to R_1, R_2 and R_3 distributed in a series circuit respectively, and V_T to the total voltage.

Current	Experimental I (mA)	$u_C(I)$ (mA)
I_1	2,210	0,048
I_2	2,210	0,048
I_3	2,210	0,048
I_T	2,210	0,048

Table 3: Individual currents where I_1, I_2 and I_3 correspond to R_1, R_2 and R_3 distributed in a series circuit, respectively, and I_T to the total current.

As can be seen in tables 2 and 3, the current's value through a series circuit is the same as the one through each resistor of said circuit. However, the voltage is different for each resistor, with the total value being the sum of individuals (table 2). This perfectly verifies the equations 3 and 4.

Finally, to verify Ohm's law we have calculated the proportionality of each resistor with its associated voltage. The results are organised in the following table.

V/R	V/R (V/Ω)	uC (V/R) (V/Ω)
$V_1/R1$	0,002223	0,000036
$V_2/R2$	0,002215	0,000048
$V_3/R3$	0,002221	0,000055
V_T/R_T	0,002228	0,000041

Table 4: Ratio between voltage and resistance in a series circuit, both individually and as a whole.

Although the ratio could be observed if we had not changed units, for a more realistic application, the different strengths and their respective uncertainties were applied in their SI unit (Ω).

Thus, we can observe that Ohm's law is indeed fulfilled given the common proportionality between the different values of resistance R and their corresponding voltage V.

3.1.3 Parallel

In a similar way to the previous section, the expected total resistance (equation 3) has been calculated so that, on comparing its value, $R = 0,2788 \pm 0,0025$ ($K\Omega$), with the one measured experimentally for the whole circuit, $R = 0,2780 \pm 0,0052$ ($K\Omega$), we can see that once again the equations agree with the results obtained.

Once more, using the DC source for individuals and total voltages and currents, we have obtained the following measures:

Voltage	Experimental V (V)	$u_C(V)$(V)
V_1	8,110	0,071
V_2	8,110	0,071
V_3	8,110	0,071
V_T	8,110	0,071

Table 5: Individual voltages where V_1, V_2 and V_3 correspond to R_1, R_2 and R_3 distributed in a parallel circuit, respectively, and V_T to the total voltage.

Current	Experimental I (mA)	$u_C(I)$ (mA)
I_1	16,72	0,16
I_2	7,980	0,094
I_3	3,690	0,059
I_T	28,60	0,53

Table 6: Individual currents where I_1, I_2 and I_3 correspond to R_1, R_2 and R_3 distributed in a parallel circuit, respectively, and I_T to the total current.

In contrast with a series circuit, in a parallel circuit, the voltage across all resistors is the same as the total voltage, as shown in Table 5; while in the case of current, the total is the sum of the current values at each of the resistors (Table 6). This was the expected behavior given the equations 6 and 7.

Finally, in order to test Ohm's law, we can compare the value of the expression $\frac{V_T}{R_T} = \frac{V_1}{R1} + \frac{V_2}{R_2} + \frac{V_3}{R3}$, again using the most commonly used unit equivalence (V/Ω).

	V/R (V/Ω)	$u_C(V/R)$ (V/Ω)
V/R_T	0,02917	0,00061
$V_1/R_1 + V_1/R_2 + V_1/R_3$	0,02909	0,00032

Table 7: Ratio between voltage and resistance in a parallel circuit.

Again, it can be seen from Table 7 that the inverse ratio described by Ohm is correct.

3.1.4 Mixed

For this circuit, described in figure 3, the expected value for the total resistance was $1,903 \pm 0,028\ k\Omega$, calculated from the equation shown in said figure. The value obtained experimentally was $1,902 \pm 0,018\ k\Omega$.

As can be seen from the proximity of the results, the theoretical expressions are correct.

3.2 Checking Ohm's Law

Finally, the circuit shown in figure 6 was constructed to supply a certain voltage to two of the five resistors so we could measure the current and test then the effectiveness of Ohm's law. For this, 10 voltage measurements were taken (between 1 mA and 10 mA) with a difference between them of about 1 mA.

3.2.1 First resistor

For R_1, the next table shows the results obtained.

Supplied V (V)	Measured V (V)	uC(V) (V)	Measured I (mA)	uC(I) (mA)
10	10,08	0,08	21,3	0,32
9	9,04	0,075	19,18	0,18
8	8,04	0,07	17,05	0,17
7	7,09	0,065	15,05	0,15
6	6,07	0,06	12,9	0,13
5	5,03	0,055	10,69	0,12
4	4,06	0,05	8,63	0,099
3	3,03	0,045	6,44	0,082
2	2,00	0,04	4,26	0,064
1	1,034	0,0082	2,19	0,048

Table 8: Measured voltage and current for each voltage value delivered to R_1, as well as their uncertainties.

And, from these values and using the least squares method we obtained the following graphic.

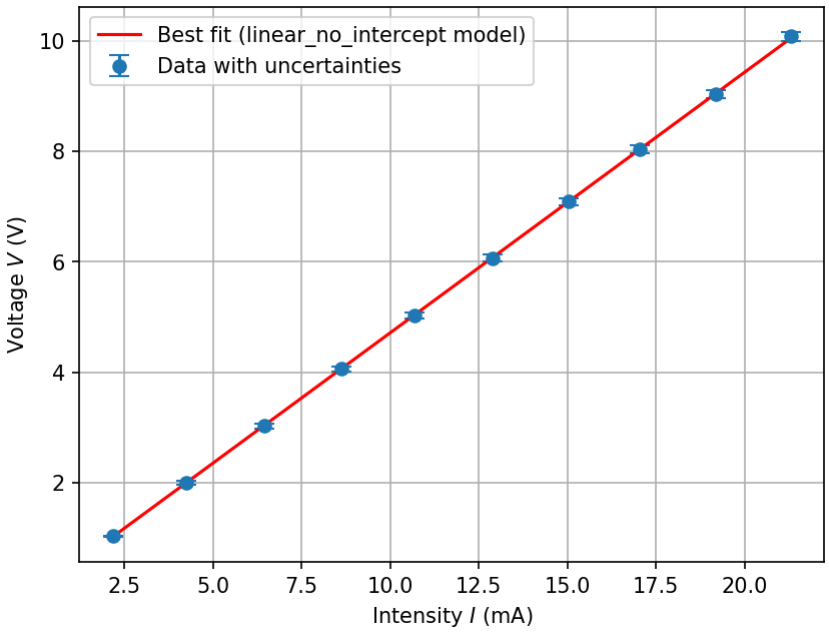


Figure 7: Voltage V *vs* current I for R_1

This adjustment has given us a linear relationship between the different values with a slope of $a = 0,470 \pm 0,001 k\Omega$, a value extremely close to the resistance of $R_1 = 0,470 \pm 0,023 k\Omega$).

Moreover, as shown in figure 7, the corresponding voltage and current values follow the proportionality of the expression we are trying to prove (1).

3.2.2 Second resistor

On the other hand, for R_2, the results obtained were the following.

Supplied V (V)	Measured V (V)	uC(V) (V)	Measured I (mA)	uC(I) (mA)
10	10,00	0,08	9,97	0,11
9	9,000	0,075	8,97	0,10
8	8,07	0,07	8,050	0,094
7	7,030	0,065	7,010	0,086
6	6,06	0,06	6,040	0,078
5	5,010	0,055	4,99	0,07
4	4,02	0,05	4,000	0,062
3	3,050	0,045	3,040	0,054
2	2,02	0,04	2,010	0,046
1	1,0110	0,0081	1,008	0,011

Table 9: Measured voltage and current for each voltage value delivered to R_2, as well as their uncertainties

Again, by applying the least squares method we obtained the graph 8.

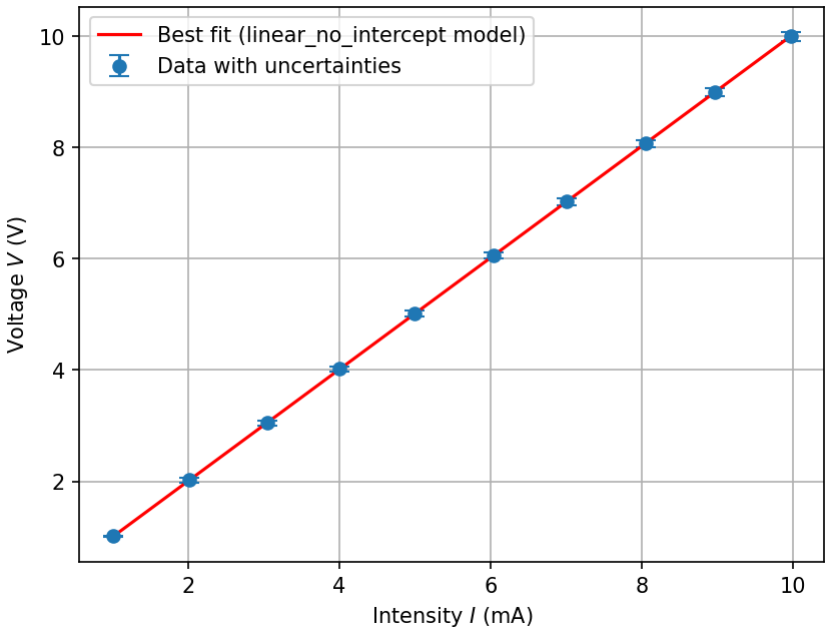


Figure 8: Voltage V over current I for R_2

As in the previous case, the slope obtained from the linear relationship between the corresponding data, $a = 1.0033 \pm 0.0031$ is very close to that of the resistor used ($R_2 = 1,00 \pm 0,05 k\Omega$).

Finally, it is visible in figure 8, that the voltage and current again follow the relation $V = IR$.

4 Conclusions

In conclusion, this experiment successfully confirmed the theoretical principles governing electrical circuits, specifically Ohm's Law and the behavior of resistors in different configurations. We were able to demonstrate that in series circuits, the total resistance is equal to the sum of the individual resistances, the voltage is divided among the resistors, and the current remains constant throughout. In contrast, for parallel circuits, the total resistance is reduced, the voltage remains the same across all resistors, and the total current is the sum of the individual branch currents.

The experimental data closely matched the theoretical predictions in both cases, with discrepancies well within the expected uncertainties. The resistors' measured values also aligned well with manufacturer specifications, reinforcing the accuracy of the experimental setup and instrumentation.

Furthermore, the linear relationship observed between voltage and current in individual resistors -confirmed through the graphical application of the least squares method - provided empirical evidence of Ohm's Law.

5 Appendixes

5.1 Calculation of Uncertainties

As we know, type A uncertainty, $u_A(x)$, which is associated to human error, can be calculated with the following equation:

$$u_A(x) = \frac{\sigma(x)}{\sqrt{N}}, \tag{8}$$

where $\sigma(x)$ represents the standard deviation and N the number of observations. On the other hand we get type B uncertainty, $u_B(x)$, related to the measuring instrument error from:

$$u_B(x) = \frac{\delta_x}{\sqrt{12}}, \tag{9}$$

where δ_x is the resolution of the measuring instrument (δ_L=0,001 m) From them, $u_C(x)$ is expressed as:

$$u_C(x) = \sqrt{u_A^2(x) + u_B^2(x)}. \tag{10}$$

Additionally, we can calculate the expanded uncertainty, which is calculated to overestimate the error. To do so, we multiplied $u_C(x)$ by a coverage factor kp that is selected for convenience:

$$U_C(x) = u_C(x)k_p. \tag{11}$$

This factor is derived from the distribution of the t-student, using $n-1$ degrees of freedom and a fixed confidence level. In this case, consistently 95%:

$$kp = 1,96.$$

So, for V_T and I_T we get:

$$u_C(V_T) = \sqrt{\left(\frac{\partial V_T}{\partial V_1}\right)^2 (u_C(V_1))^2 + \left(\frac{\partial V_T}{\partial V_2}\right)^2 (u_C(V_2))^2 + \left(\frac{\partial V_T}{\partial V_3}\right)^2 (u_C(V_3))^2} = \tag{12}$$

$$= \sqrt{(u_C(V_1))^2 + (u_C(V_2))^2 + (u_C(V_3))^2} \ \ (V)$$

$$u_C(I_T) = \sqrt{\left(\frac{\partial I_T}{\partial I_1}\right)^2 (u_C(I_1))^2 + \left(\frac{\partial I_T}{\partial I_2}\right)^2 (u_C(I_2))^2 + \left(\frac{\partial I_T}{\partial I_3}\right)^2 (u_C(I_3))^2} = \tag{13}$$

$$= \sqrt{(u_C(I_1))^2 + (u_C(I_2))^2 + (u_C(I_3))^2} \ (A)$$

Regarding the function V/R, which we will call f, its uncertainty is

$$u_C(V/R) = \sqrt{\left(\frac{\partial f}{\partial V}\right)^2 (u_C(V))^2 + \left(\frac{\partial f}{\partial R}\right)^2 (u_C(R))^2} = \tag{14}$$

$$= \sqrt{\left(\frac{1}{R}\right)^2 (u_C(V))^2 + \left(\frac{-V}{R^2}\right)^2 (u_C(R))^2} \ (A)$$

Now, we will call the sum $\frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$, Sum. Its the uncertainty will be

$$u_C(Sum) = \sqrt{\left(\frac{\partial Sum}{\partial(V/R_1)}\right)^2 (u_C(V/R_1))^2 + \left(\frac{\partial Sum}{\partial(V/R_2)}\right)^2 (u_C(V/R_2))^2 + \left(\frac{\partial Sum}{\partial(V/R_3)}\right)^2 (u_C(V/R_3))^2} \tag{15}$$

$$= \sqrt{(u_C(V/R_1))^2 + (u_C(V/R_2))^2 + (u_C(V/R_3))^2}(A)$$

For R_T in the series circuit built with R_1, R_2, R_3, we get the uncertainty

$$u_C(R_T) = \sqrt{\left(\frac{\partial R_T}{\partial R_1}\right)^2 (u_C(R_1))^2 + \left(\frac{\partial R_T}{\partial R_2}\right)^2 (u_C(R_2))^2 + \left(\frac{\partial R_T}{\partial R_3}\right)^2 (u_C(R_3))^2} \tag{16}$$

$$= \sqrt{(u_C(R_1))^2 + (u_C(R_2))^2 + (u_C(R_3))^2}(k\Omega)$$

In contrast, for R_T in the parallel circuit studied, as we can simplify $R_T = \frac{R_1R_2R_3}{R_2R_3+R_1R_3+R_1R_2}$, the are looking for is

$$u_C(R_T) = \sqrt{\left(\frac{\partial R_T}{\partial R_1}\right)^2 (u_C(R_1))^2 + \left(\frac{\partial R_T}{\partial R_2}\right)^2 (u_C(R_2))^2 + \left(\frac{\partial R_T}{\partial R_3}\right)^2 (u_C(R_3))^2}(k\Omega) \tag{17}$$

where

$$\frac{\partial R_T}{\partial R_1} = R_2R_3\frac{(R_2R_3 + R_1R_3 + R_1R_2) - R_1(R_2 + R_3)}{(R_2R_3 + R_1R_3 + R_1R_2)^2} \tag{18}$$

$$\frac{\partial R_T}{\partial R_2} = R_1R_3\frac{(R_2R_3 + R_1R_3 + R_1R_2) - R_2(R_1 + R_3)}{(R_2R_3 + R_1R_3 + R_1R_2)^2} \tag{19}$$

$$\frac{\partial R_T}{\partial R_3} = R_1R_2\frac{(R_2R_3 + R_1R_3 + R_1R_2) - R_3(R_1 + R_2)}{(R_2R_3 + R_1R_3 + R_1R_2)^2} \tag{20}$$

Finally, R_T in the mixed association from figure 3, we will calculate the corresponding uncertainties of $u_C(R_6)$ and $u_C(R_7)$

$$u_C(R_6) = \sqrt{\left(\frac{\partial R_6}{\partial R_1}\right)^2 (u_C(R_1))^2 + \left(\frac{\partial R_6}{\partial R_2}\right)^2 (u_C(R_2))^2 + \left(\frac{\partial R_6}{\partial R_3}\right)^2 (u_C(R_3))^2} \tag{21}$$

$$= \sqrt{\left(\frac{R_2^2}{(R_1+R_2)^2}\right)^2 (u_C(R_1))^2 + \left(\frac{R_1^2}{(R_1+R_2)^2}\right)^2 (u_C(R_2))^2 + (u_C(R_3))^2}(k\Omega)$$

$$u_C(R_7) = \sqrt{\left(\frac{\partial R_7}{\partial R_4}\right)^2 (u_C(R_4))^2 + \left(\frac{\partial R_7}{\partial R_5}\right)^2 (u_C(R_5))^2} = \sqrt{(u_C(R_4))^2 + (u_C(R_5))^2}(k\Omega) \tag{22}$$

and, with them, R_T's uncertainty

$$u_C(R_T) = \sqrt{\left(\frac{\partial R_T}{\partial R_6}\right)^2 (u_C(R_6))^2 + \left(\frac{\partial R_T}{\partial R_7}\right)^2 (u_C(R_7))^2} \tag{23}$$

$$= \sqrt{\left(\frac{R_7^2}{(R_6+R_7)^2}\right)^2 (u_C(R_6))^2 + \left(\frac{R_6^2}{(R_6+R_7)^2}\right)^2 (u_C(R_7))^2}(k\Omega)$$

References

[1] Irene Adroher Benítez et al. *Basic Experimental Techniques Laboratory Manual*. Ed. by Raúl Rica Alarcón. University of Granada, 2025.